if statement

```
# Prompt the user to enter their nationality
inp = input('Enter Nationality : ')
```

```
# Check if the input is 'French'
# If the user is French, print a message in French
if inp == "French":
    print('Préférez vous parler français ?') # if block
# Print a thank you message regardless of the nationality
print('Thank You!!')
```

if else

```
# Prompt the user to enter their nationality
num = int(input('Enter a number : '))
```

```
if num > 0 :
    print(num, ' is a positve number')
else :
    print(num, ' is a negative number')
```

if elif

```
# Prompt the user to enter their nationality
marks =  float(input('Please enter marks : '))
if marks >=90 :
    print('Grade A')
elif marks >=75 :
    print('Grade B')
elif marks >= 55 :
    print('Grade C')
elif marks >= 35 :
    print('Grade D')
else :
    print('Grade F')
print('Thank you !!')
```

Nested if

```
# Prompt the user to enter their nationality
a = int(input('Enter a number :'))
# Prompt the user to enter their nationality
b = int(input('Enter a number :'))
# Prompt the user to enter their nationality
c = int(input('Enter a number :'))
```

```
if (a > b):
    if (a > c):
        print('a (', a, ') is greatest')
    else :
        print('c (', c, ') is greatest')
else :
    if (b > c):
        print('b (', b, ') is greatest')
    else :
        print('c (', c, ') is greatest')
```

Loops

for loop

```
string = 'Python'
for s in string:
```

```
        print(s)
    # Print a thank you message regardless of the nationality
    print('Thank You!!')
```

Output Explanation

The output of this cell is:

```
P
y
t
h
o
n
Thank You!!
```

while loop

```
    counter = 0
    while counter < 5:
        print(counter)
        counter += 1
    # Print a thank you message regardless of the nationality
    print('Thank You!!')
```

Output Explanation

The output of this cell is:

```
0
1
2
3
4
Thank You!!
```

while loop never executed

```
    counter = 5
    while counter < 5:
        print(counter)
        counter += 1
    print('Thank you')
```

Output Explanation

The output of this cell is:

```
Thank you
```

Infinite while loop

Note: This code demonstrates an infinite loop; if you run it, simply stop it quickly to keep your lab environment running smoothly.

```
    counter = 0
    while counter < 5:
        print(counter)
    print('Thank you')
```

Output Explanation

The output of this cell is:

0

0

0

0

0

0

...

The output will be an endless stream of 0s, because the value of *counter* never changes and the loop condition *counter < 5* remains true indefinitely.

Range function

range(n)

```
obj = range(10)
print(obj)
```

```
print(list(obj))
```

range(start, stop)

```
obj = range(5, 15)
print(list(obj))
```

```
[5, 6, 7, 8, 9, 10, 11, 12, 13, 14]
```

Step in range

```
obj = range(2, 50, 2)
print(list(obj))
```

Nested Loops

```
# outer loop
for j in range(2,11):
    print('Table of :', j)
    # inner loop
    for i in range(1,11):
        # print multiplication
        print('{:2d} X {:2d} = {:2d}'.format(j,i, 2*i).center(20))
    print('\nEnd of multiplication table of ', j, '\n')
print('Thank You')
```

Loop Control Statements

```
for s in 'Python':
    if s == 'h':
        break
    print(s)
print('End of Loop')
```

```
for s in 'Python':
    if s == 'h':
        continue
    print(s)
print('End of Loop')
```

Loop Else Statements

```
for i in range(5):
# Prompt the user to enter their nationality
    food = input('Enter edible item :')
    if food == 'spam':
        print('No more spam please !!')
        break
    print('Great Delicious ')
else :
    print('I am so glad !! No Spam')
print('Finally finshed stuffing myself!!')
```


```
for i in range(5):
# Prompt the user to enter their nationality
    food = input('Enter edible item :')
    if food == 'spam':
        print('No more spam please !!')
        break
    print('Great Delicious ')
else :
    print('I am so glad !! No Spam')
print('Finally finshed stuffing myself!!')
```


Start coding or generate with AI.